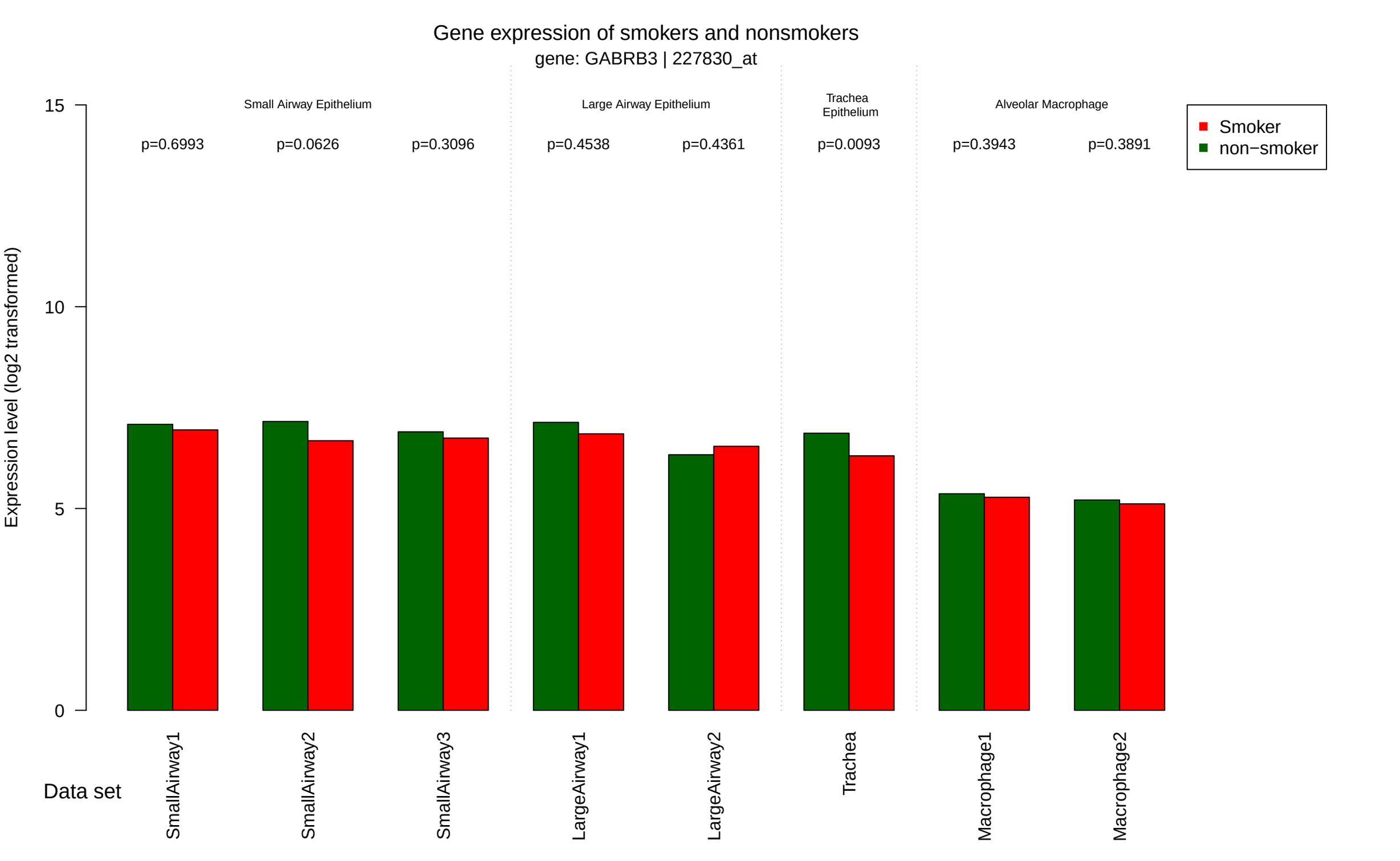
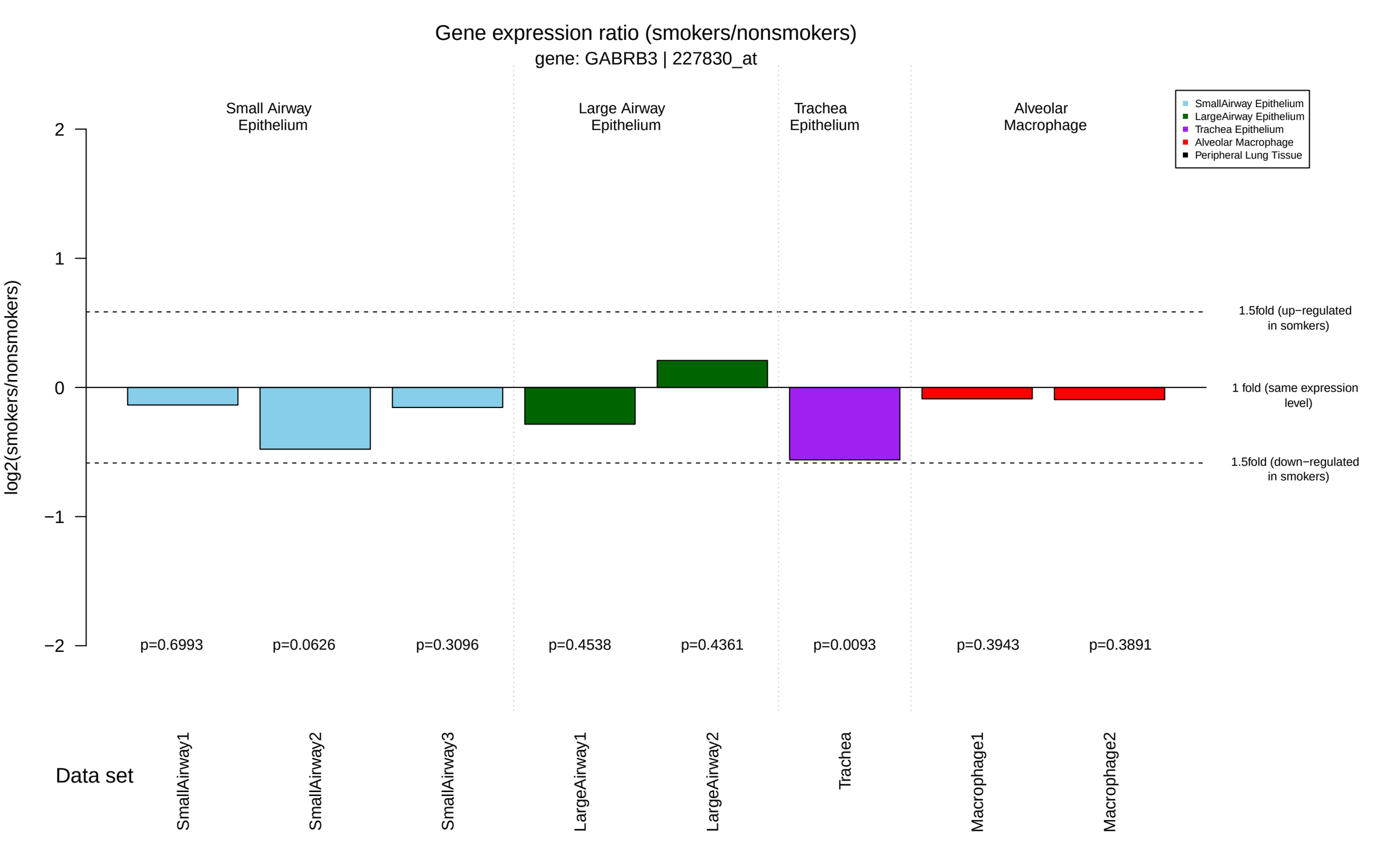
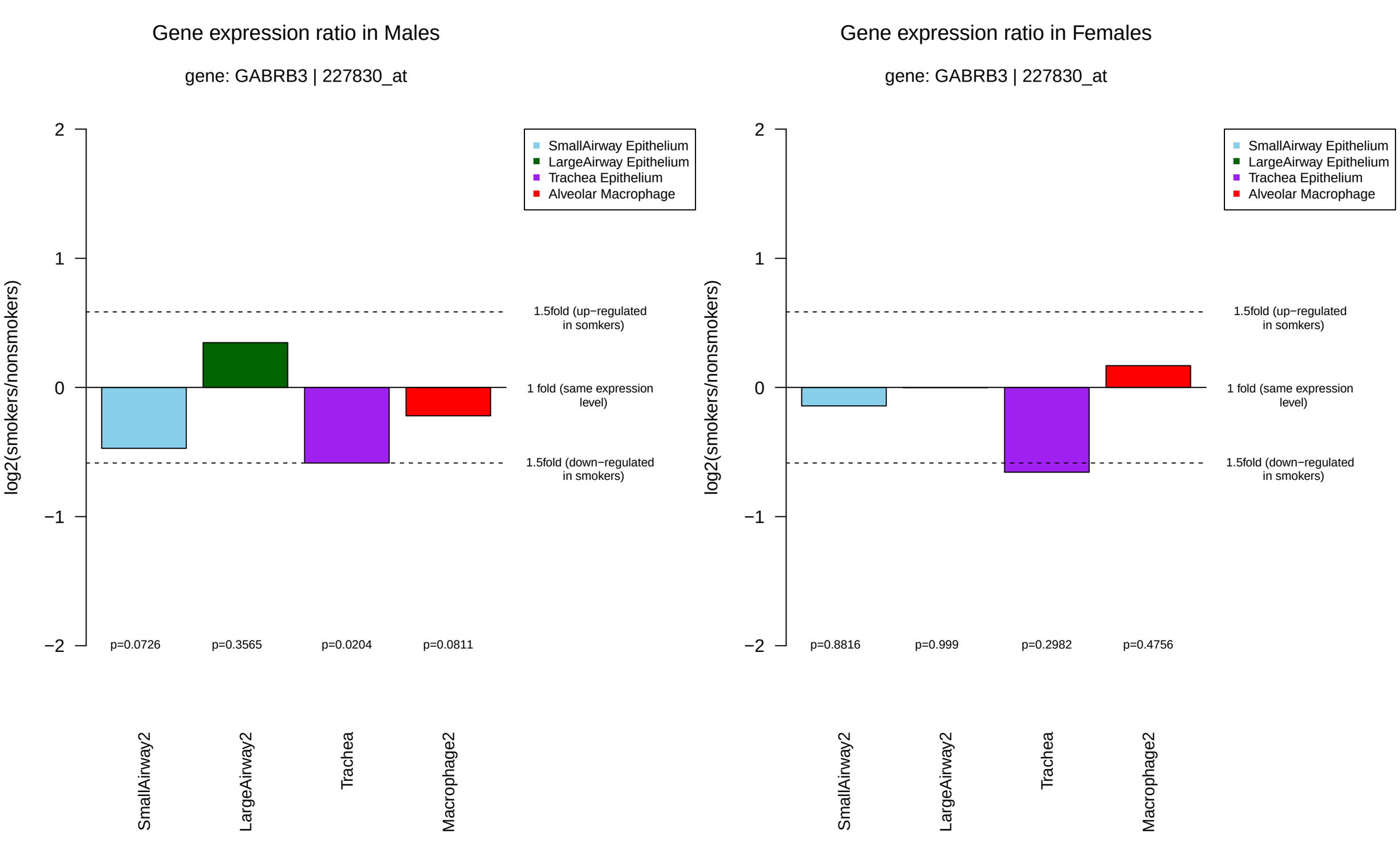
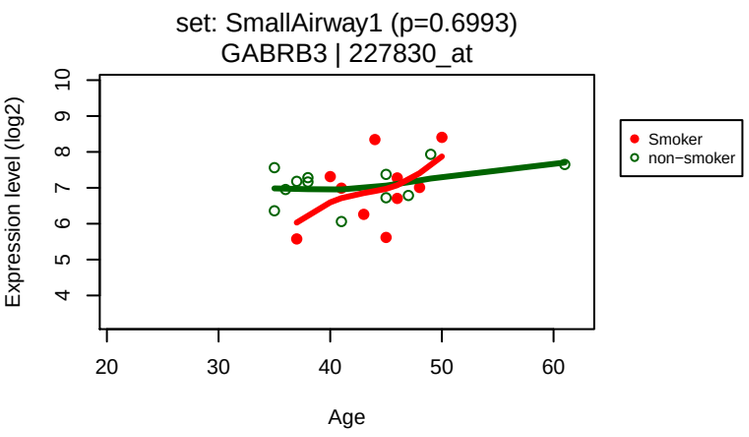
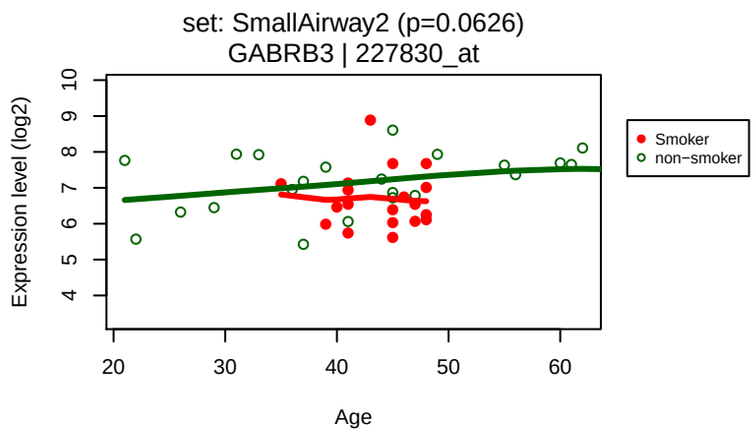
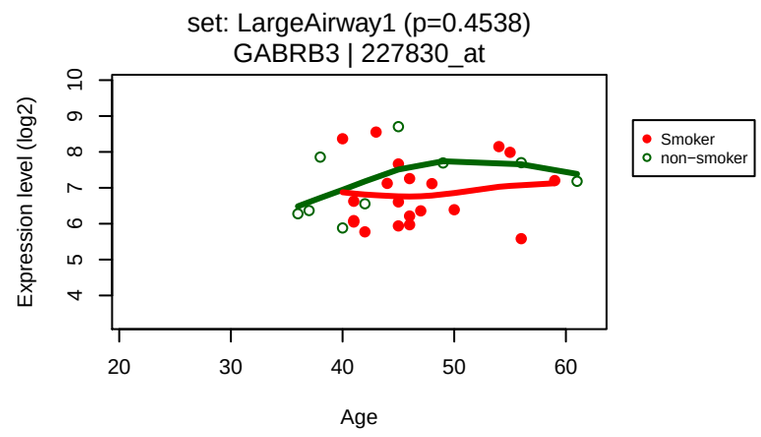

Gene expression of smokers and nonsmokers
gene: GABRB3 | 227830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6993
p=0.0626
p=0.3096
p=0.4538
p=0.4361
p=0.0093
p=0.3943
p=0.3891
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

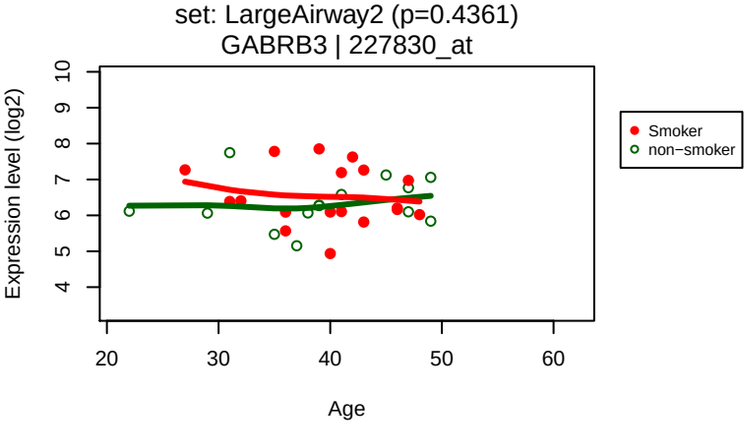

Gene expression ratio (smokers/nonsmokers)
gene: GABRB3 | 227830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6993
p=0.0626
p=0.3096
p=0.4538
p=0.4361
p=0.0093
p=0.3943
p=0.3891
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

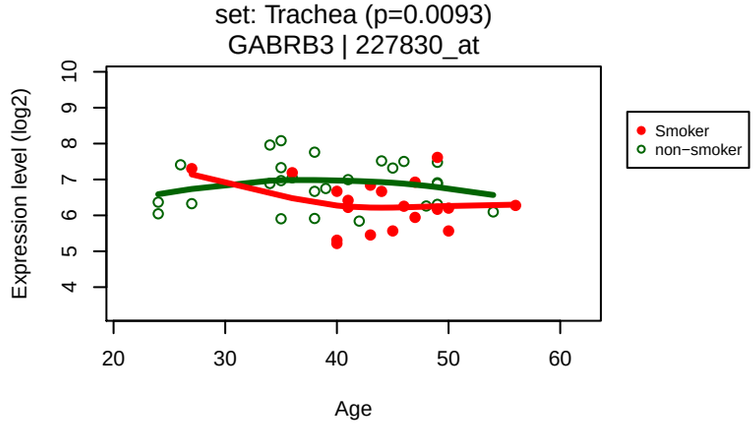

Gene expression ratio in Males
gene: GABRB3 | 227830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0726
p=0.3565
p=0.0204
p=0.0811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRB3 | 227830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8816
p=0.999
p=0.2982
p=0.4756
SmallAirway2
LargeAirway2
Trachea
Macrophage2

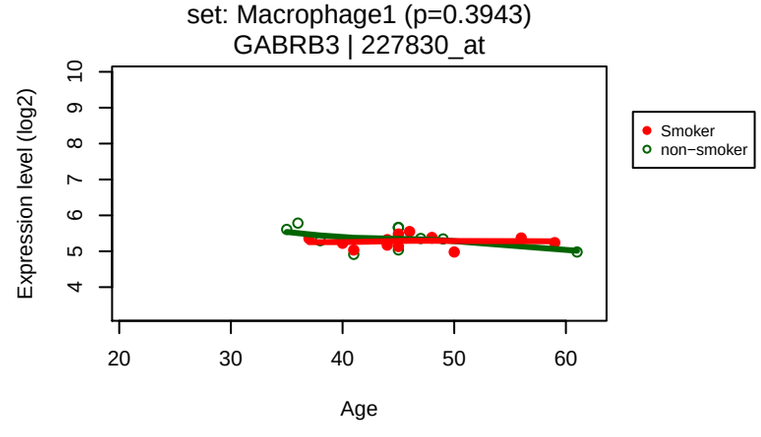

set: SmallAirway1 (p=0.6993)
GABRB3 | 227830_at
Expression level (log2)
Age
Smoker
non−smoker

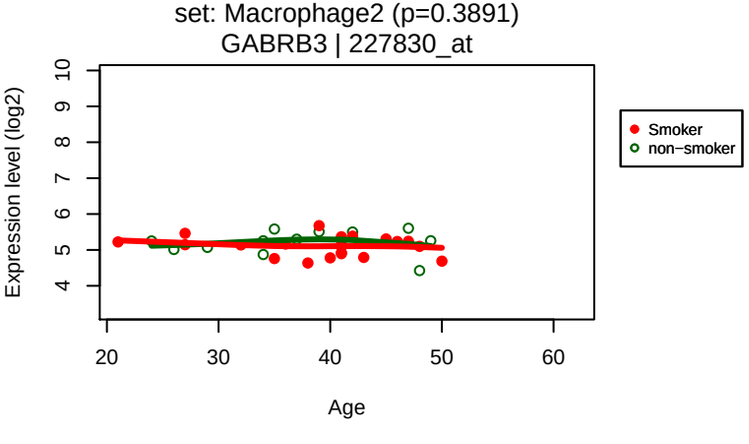

set: SmallAirway2 (p=0.0626)
GABRB3 | 227830_at
Expression level (log2)
Age
Smoker
non−smoker

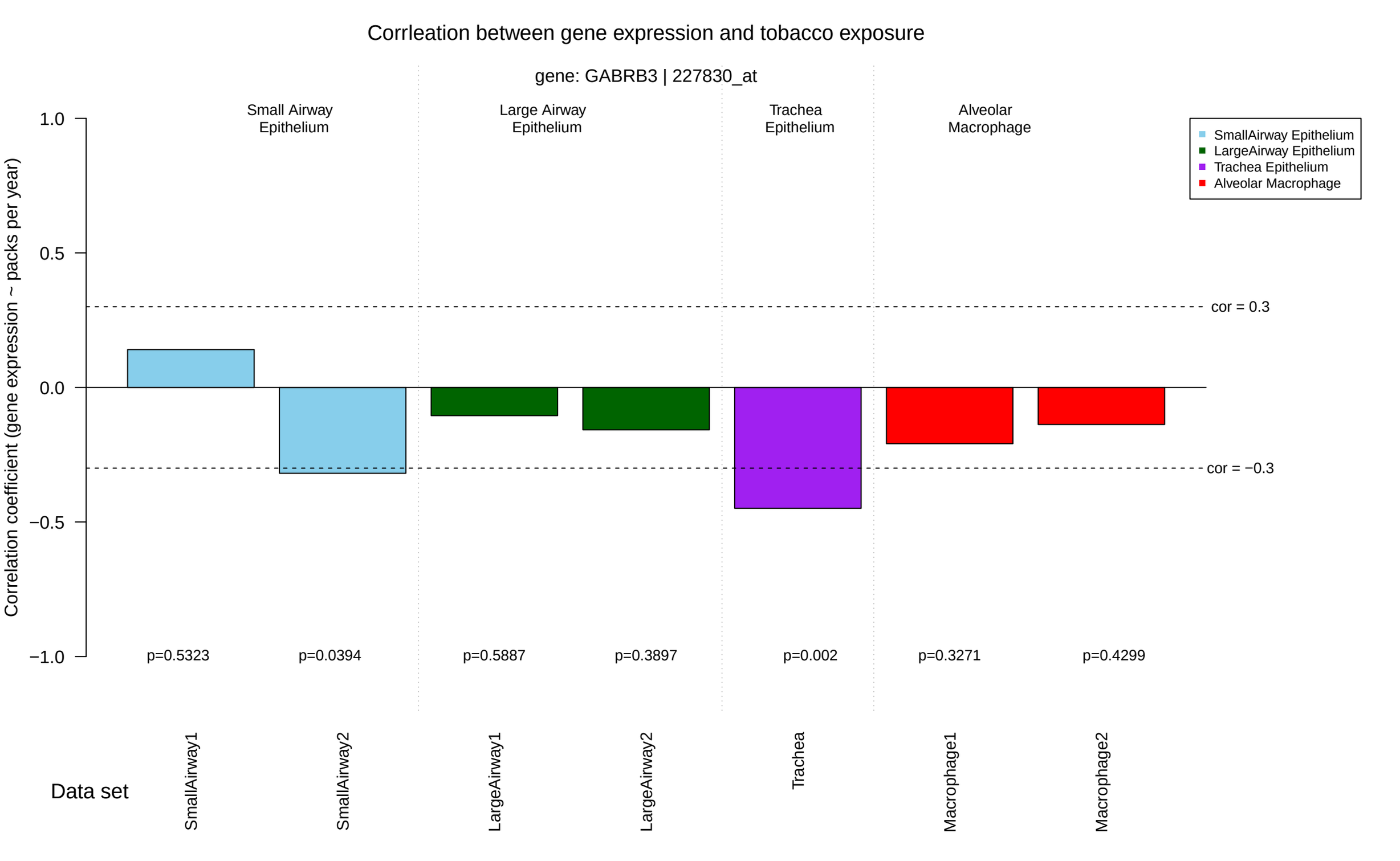

set: LargeAirway1 (p=0.4538)
GABRB3 | 227830_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4361)
GABRB3 | 227830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0093)
GABRB3 | 227830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3943)
GABRB3 | 227830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3891)
GABRB3 | 227830_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRB3 | 227830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5323
p=0.0394
p=0.5887
p=0.3897
p=0.002
p=0.3271
p=0.4299
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2